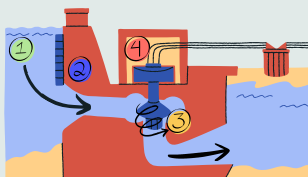


Hydroelectric Power

HOW IT WORKS



1 Water Source
(reservoir or river provides flowing water)

2 Dam
(water is gathered in dams)

3 Turbine Rotation
(flowing water spins the turbines)

4 Generator
(moving turbines generate electricity)

TYPES OF HYDROELECTRIC SYSTEMS

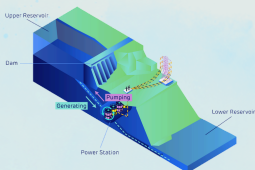


1. Impoundment

- Uses a dam to block a river and create a reservoir
- Stored water is released and flows downward through turbines, generating electricity
- Allows controlled and reliable power generation

2. Diversion (Run-of-River)

- Channels part of a river away from its natural path
- Uses natural elevation drop rather than a large dam
- Water flows through a penstock to spin turbines and generate electricity



3. Pumped Storage Hydropower (PSH)

- Uses two reservoirs at different heights
- During low electricity demand, excess energy pumps water uphill
- During high demand, water flows downhill to generate electricity

ADVANTAGES

They have a longer lifespan & are cost efficient

They are more reliable than other forms of renewable energy

They are capable of producing large amounts of energy and storing excess energy

They can have multiple additional uses, for example acting as a bridge for traffic or connecting power/communication cables



Renewable and sustainable: Unlike coal and oil and natural gas, water flow is naturally replenished

DISADVANTAGES

- They can cause major flooding of natural areas, habitats, and riverside settlements
- They are often very expensive to set up, ranging from a few million dollars for smaller, local projects to upwards of billions of dollars for more massive, complex structures
- They can take many years to fully construct
- Many people believe that dams are visually polluting and that they spoil the view in places of previous natural beauty

